

Mahi Singh

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Education

Vellore Institute of Technology

May 2027

B.Tech, Computer Science and Engineering — CGPA: **9.20/10**

Mercy Memorial School

2023 & 2021

ISC (Class XII): **89.75%** | ICSE (Class X): **95.40%**

Technical Skills

- **Languages:** Python, JavaScript, SQL
- **AI & Conversational Systems:** LLM APIs (Gemini 2.5, Groq / Llama 3.3), Prompt Engineering, RAG & Vector Search (Supabase pgvector), Speech Recognition (Web Speech API)
- **Backend & APIs:** FastAPI, Node.js / Express, REST API Design, PostgreSQL / Supabase
- **Tools:** Git, GitHub, VS Code

Experience

AI Engineer Intern – AmasQIS.ai (Remote)

Mar 2025 – Sep 2025

- Built and deployed a real-time PPE compliance detection pipeline (Python, TensorFlow) processing live multi-camera factory feeds at 90%+ precision
- Optimized model inference for low-latency production deployment through structured build-test-iterate cycles with cross-functional engineering teams

Projects

Saathi – Voice Command Shopping Assistant

[\[GitHub\]](#)

React, Node/Express, Groq LLM

- Designed a mobile-first voice shopping UI with live transcription, animated mic feedback, and category-based list organization
- Built NLP-driven voice search supporting 4 filter types (brand, price, size, organic) with compound-filter parsing, verified across 20 queries and 8 filter combinations
- Used a RICE-style reach/impact lens to prioritize a critical filter-matching bug causing zero search results despite accurate intent parsing – fixed the root cause, validating a 75% real match rate

KumbhSaathi – Multilingual Pilgrim Assistant

[\[GitHub\]](#)

Python, FastAPI, Groq (Llama 3.3), Web Speech API

- Built a multilingual AI chat assistant (Hindi, English, Bengali, Tamil, auto-detected) with voice input, for 100M+ Mahakumbh 2028 pilgrims
- Designed a hybrid query pipeline – sub-100ms local FAQ matching before falling back to Groq's Llama 3.3 70B – cutting unnecessary LLM calls
- Added live family-location sharing (group PIN + GPS map) and one-tap emergency SOS with real GPS coordinates and helpline numbers

JeevanTrack – AI Health Timeline Platform

[\[GitHub\]](#)

Next.js, FastAPI, Supabase, Gemini 2.5

- Built an AI platform using Gemini Vision to extract and summarize medical reports into a unified timeline, cutting manual data entry by 80%
- (*In progress*) Building a two-tier RAG pipeline for “Ask AI”: semantic retrieval via Gemini Embedding-2 over Supabase pgvector, with SQL fallback
- Designed retrieval to resolve fuzzy, non-keyword health questions that plain SQL search fails on, by embedding normalized report fields at upload

Achievements

- **Mahakumbh Innovation Hackathon 2028:** Ranked 8th domain-wise (Product track) building KumbhSaathi; short-listed for Expert Hire's Special Placement Drive
- **InnovateZ 2026 (The Gen Z Challenge):** Advanced to Round 2, ranking in the top 60 of 700+ teams; delivered a working MVP with live demo and final presentation

Volunteering

- **eVidyaloka:** Remote educator, 2 years teaching mathematics to 50+ rural Madhya Pradesh students, improving assessment scores by 30%
- **Pahal Foundation:** Tutored construction workers' children; drove formal-school enrollment by identifying high-potential students and persistently engaging skeptical parents